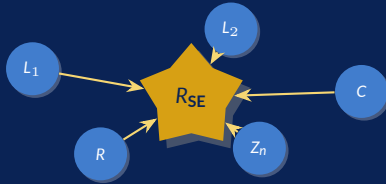

FINAL PRESENTATION

Direct Optimization of the RLC Circuit Components of an
Intelligent Reflecting Surface (IRS) using Memetic PSO

Resonant-circuit model — algorithm design — Monte-Carlo simulation



- 1 Context and final problem statement
- 2 Physical constraints and the origin of every bound
- 3 Circuit-model verification
- 4 Algorithm design: from plain PSO to memetic PSO
- 5 Simulation results
- 6 Complexity and conclusion

- 1 Context and final problem statement
- 2 Physical constraints and the origin of every bound
- 3 Circuit-model verification
- 4 Algorithm design: from plain PSO to memetic PSO
- 5 Simulation results
- 6 Complexity and conclusion

Weeks 1–4 (completed)

Optimized the **phase vector** $\theta \in [-\pi, \pi]^N$ on the equivalent phase-shift model, Eq. (5) of the reference paper (arXiv:1907.06002).

- Clerc–Kennedy constriction PSO
- Compared against the paper's AO

Final requirement

Directly optimize the physical circuit components $\{L_{1,n}, L_{2,n}, C_n, R_n\}_{n=1}^N$ of every IRS element – in the spirit of the *Electronics for IT* course.

$L_{1,n}, L_{2,n}, C_n, R_n$

Z_n (impedance)

v_n (reflection coeff.)

R_{SE} (rate)

Communication system

- AP with $M = 2$ antennas; IRS with $N = 40$ elements 500 m from the AP; user at distance d
- Rayleigh fading with distance-dependent path loss; $P_t = 36$ dBm, $\sigma^2 = -94$ dBm

$$R_{SE} = \log_2 \left(1 + \frac{P_t \|\mathbf{v}^H \Phi + \mathbf{h}_d^H\|^2}{\sigma^2} \right)$$

Resonant circuit per element

$$Z_n = \frac{j\omega L_{1,n}(j\omega L_{2,n} + \frac{1}{j\omega C_n} + R_n)}{j\omega L_{1,n} + j\omega L_{2,n} + \frac{1}{j\omega C_n} + R_n}$$

$$v_n = \frac{Z_n - Z_0}{Z_n + Z_0}, \quad Z_0 = 377 \Omega$$

- $\omega = 2\pi \times 2.4$ GHz (ISM band)
- $|v_n| \leq 1$: passive circuit, $R_n > 0$

Final optimization problem

$\max_{\{L_{1,n}, L_{2,n}, C_n, R_n\}} R_{SE}$ subject to box constraints on every component – a

$D = 4N = 160$ -dimensional search space; the Eq. (5) phase model is **no longer used**.

- 1 Context and final problem statement
- 2 Physical constraints and the origin of every bound**
- 3 Circuit-model verification
- 4 Algorithm design: from plain PSO to memetic PSO
- 5 Simulation results
- 6 Complexity and conclusion

Component bounds and their origins

Variable	Range	Origin
C_n	[0.47, 2.35] pF	Exactly the paper's varactor tuning range (SMV1231, 0–15 V); the 5:1 ratio is the practical limit of silicon varactors at 2.4 GHz
$L_{1,n}$	[0.5, 5.0] nH	Brackets the nominal 2.5 nH; a <i>design freedom</i> of the element geometry – lets PSO retune each element's resonance
$L_{2,n}$	[0.1, 2.0] nH	Brackets the nominal 0.7 nH (top-layer/parasitic inductance)
R_n	[0.5, 5.0] Ω	Paper measures {1, 2.5} Ω ; the 0.5 Ω floor = diode junction + PCB dielectric + skin-effect losses

Why not widen the bounds to “cheat” the rate upward?

A 50:1 varactor does not exist at 2.4 GHz; below ≈ 0.3 pF layout parasitics dominate; $R < 0.5 \Omega$ implies a nearly lossless circuit – unphysical. **Only with realistic bounds does the comparison carry engineering meaning.**

- 1 Context and final problem statement
- 2 Physical constraints and the origin of every bound
- 3 Circuit-model verification**
- 4 Algorithm design: from plain PSO to memetic PSO
- 5 Simulation results
- 6 Complexity and conclusion

Exp. 1 – The circuit model reproduces the paper's phase model

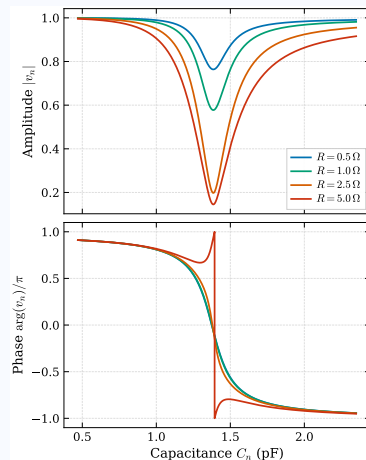
- Fix $L_1 = 2.5 \text{ nH}$, $L_2 = 0.7 \text{ nH}$, $R = 2.5 \Omega$; sweep C over the varactor range
- Fit the Eq. (5) constants to the resulting (amplitude, phase) curve:

Fit result (RMSE = 0.0029)

$$(\beta_{\min}, k, \phi)_{\text{circuit}} = (0.21, 1.75, 0.435\pi)$$

$$(\beta_{\min}, k, \phi)_{\text{paper}} = (0.2, 1.6, 0.43\pi)$$

The empirical constants $\beta_{\min}, k, \phi$ are **derivable** from the circuit – the computation chain is verified before any optimization.

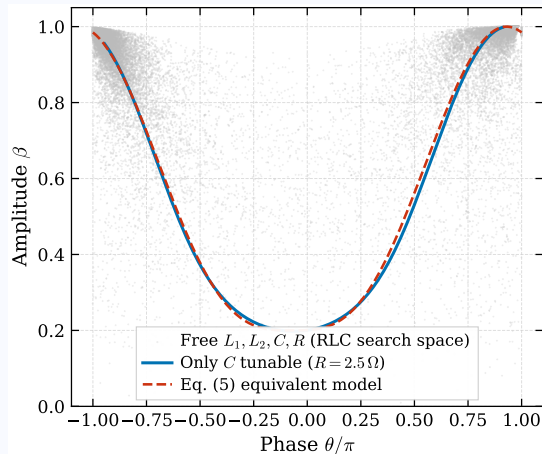


Why does component optimization beat phase optimization?

- **Only C tuned** (the paper's problem): the (phase, amplitude) pair is *locked onto a single curve* $\beta(\theta)$ — each desired phase dictates its amplitude
- **All four components free**: the same phase is reachable by many component combinations with different amplitudes — the feasible set inflates from a *curve* into a *region*
- PSO picks, for every required phase, the combination with the **highest amplitude**

Key insight

This is the fundamental reason component-level optimization beats AO — which is confined to the Eq. (5) curve.



- 1 Context and final problem statement
- 2 Physical constraints and the origin of every bound
- 3 Circuit-model verification
- 4 Algorithm design: from plain PSO to memetic PSO**
- 5 Simulation results
- 6 Complexity and conclusion

Normalized encoding

- Wildly different scales: $L \sim 10^{-9}$, $C \sim 10^{-12}$, $R \sim 10^0$
- PSO particle $\mathbf{u} \in [0, 1]^D$, $D = 4N$; linear decode to physical ranges before each evaluation
- Single $v_{\max} = 0.2$ for all 160 dimensions; the whole swarm is evaluated in one vectorized NumPy call

Base PSO (inherited from week 4)

- Clerc-Kennedy constriction:
 $w = 0.7298$, $c_1 = c_2 = 2.05$
- APSO-style escape: single-dimension jump on gbest with decaying amplitude
 $\sigma = 0.5 (1 - t/T)$

Three anomalies of the first version (fixed budget)

(1) The rate-vs- N curve **plateaued** at $N = 70-80$; (2) the “**fewer variables win**” paradox: all-four-free (3.93) lost to C, R -only (3.97) — mathematically impossible; (3) single-run convergence below the Monte-Carlo mean.

Diagnosis: curse of dimensionality — the same 60 particles $\times$ 300 iterations for problems from 40 to 320 dimensions $\Rightarrow$ a *search-resource allocation* error, not a physics error.

Four improvements — a five-version journey

1. **Dimension-adaptive budget:** $n_p = \max(60, \min(120, \lfloor 0.6D \rfloor))$,
 $T = \max(400, \min(800, 4D)) \Rightarrow$ the paradox disappeared, inclusion ordering restored
2. **Unlocking 1 to 4 variables per element:** the theoretically correct ordering emerged: $C < C, L_1 < C, R < \text{all four}$
3. **Phase-aligned warm start:** seed particle aligned to ideal phases via a 256-point grid of $C \Rightarrow 3.942 \rightarrow 3.996$ bps/Hz (10 trial channels)
4. **Local L-BFGS-B polishing (memetic):** R_{SE} is differentiable in all four variables $\Rightarrow$ after PSO, quasi-Newton climbs straight to the basin's peak

Warm start $\rightarrow$ PSO (T iters, global exploration) $\rightarrow$ **L-BFGS-B** (≤ 160 iters, local polishing)

A lesson about benchmarks

PSO-RLC (2.620) once “beat” the ideal upper bound (2.558) — because the bound itself was under-converged (5 sweeps). Raising to 30 sweeps: $2.558 \rightarrow 2.772$. **When the algorithm improves, re-validate the benchmarks.**

- **12-core parallel Monte Carlo**
(ProcessPoolExecutor, reproducible seeds): one 500-realization point drops from ≈ 15 min to ≈ 1.7 min ($\approx 9\times$)
- **Reducing realizations** 1000 $\rightarrow$ 500: standard error of the mean $\sigma/\sqrt{500} \approx 0.012$ bps/Hz ($< 0.5\%$); 1000 realizations would only reach 0.009 — not worth double the runtime
- **Pinning** $d = 498$ m for the comparison experiments — the exact point of the paper's Fig. 6 (user closest to the IRS)

	Sequential	12 cores
One 500-run point	≈ 15 min	≈ 1.7 min
Full 5-exp. suite	6–7 h	54 min

Scale

$\approx 13,500$ memetic optimization runs in the final experiment suite.

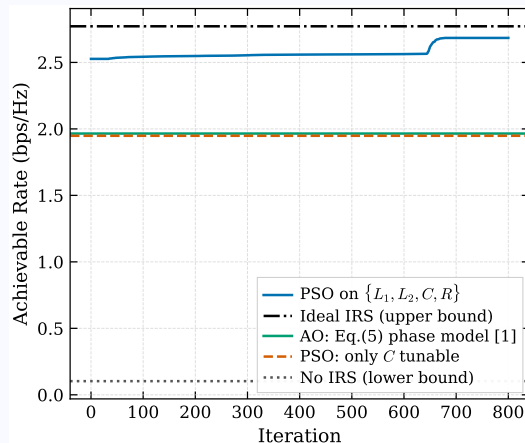
- 1 Context and final problem statement
- 2 Physical constraints and the origin of every bound
- 3 Circuit-model verification
- 4 Algorithm design: from plain PSO to memetic PSO
- 5 Simulation results**
- 6 Complexity and conclusion

Exp. 2 – Convergence on a single fixed channel

Scheme	bps/Hz	% of ideal
Ideal IRS	2.772	100%
Memetic PSO (4 vars)	2.684	96.8%
AO on Eq. (5)	1.965	70.9%
PSO, only C	1.948	70.3%
No IRS	0.103	—

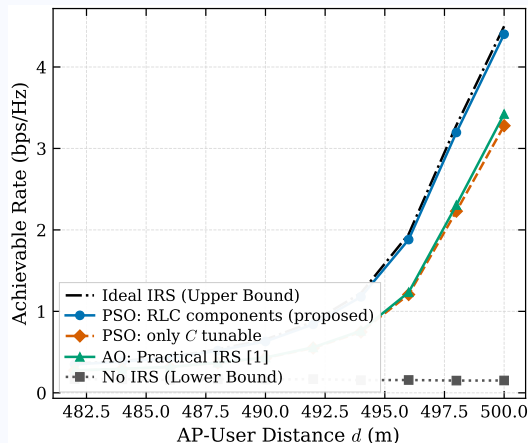
Cross-validation

PSO-C-only (1.948) $\approx$ AO (1.965): two entirely different implementations converge to the same level – confirming both the circuit model and the pipeline.



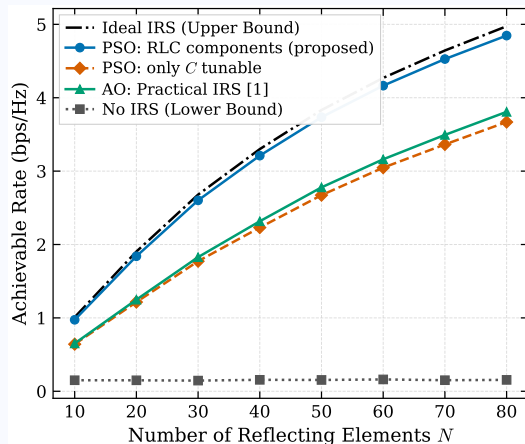
Exp. 3 – Rate versus AP-user distance

- 500 Monte-Carlo realizations per point, $d = 482\text{--}500\text{ m}$
- At $d = 498\text{ m}$: PSO-RLC reaches **3.196 bps/Hz** (97.3% of the ideal bound 3.284), exceeding AO (2.305) by **38.7%**
- Every IRS-assisted curve **rises monotonically** as the user approaches the IRS; the “no IRS” line stays flat at $\approx 0.15\text{ bps/Hz}$
- Large gap between $d = 498\text{ }(\approx 3.2)$ and $d = 500\text{ }(\approx 4.4)$: the system is highly sensitive to the user-IRS distance (2 m vs. 2.83 m)



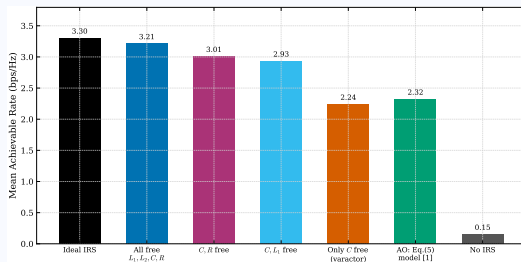
Exp. 4 – Rate versus the number of elements N

- Setup of the paper's Fig. 6, $d = 498$ m, $N = 10$ –80
- $N = 40$: PSO-RLC reaches **3.211 bps/Hz** (97.3% of the ideal bound)
- $N = 80$ – a **320-dimensional** search space – still holds 97.4% of the ideal bound (4.847 vs. 4.974)
- Thanks to the adaptive budget, the PSO-RLC curve **keeps rising up to $N = 80$** – the v0 plateau is gone



Exp. 5 – Influence of each component (the central experiment)

Configuration	Mean	Std.	% of ideal
Ideal IRS (upper bound)	3.301	0.279	100.0%
All four free: L_1, L_2, C, R (proposed)	3.211	0.277	97.3%
C, R free	3.006	0.276	91.1%
C, L_1 free	2.932	0.277	88.8%
Only C free (varactor)	2.239	0.294	67.8%
AO: Eq. (5) phase model	2.320	0.279	70.3%
No IRS (lower bound)	0.150	0.102	4.6%

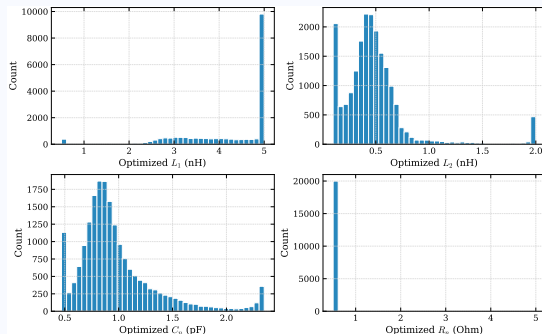


($d = 498$ m, $N = 40, 500$ Monte-Carlo realizations)

- C_n is the foundational variable: alone it reaches $2.239 \approx$ AO (2.320) – reconfirming the circuit/Eq. (5) equivalence
- Adding R_n (3.006) beats adding $L_{1,n}$ (2.932): **cutting loss is the most direct improvement channel**
- All four free: **3.211 bps/Hz = 97.3% of the ideal bound**; the remaining 0.090 bps/Hz is *irreducible physical loss* ($R_n \geq 0.5 \Omega$)

Optimized component distributions – hardware meaning

- R_n **piles up against its lower bound**
 $0.5\ \Omega$: the optimizer “understands” that loss only hurts $\Rightarrow$ investing in low-loss diodes/materials pays off most
- $L_{1,n}, L_{2,n}, C_n$ **spread across the interior**: each element picks its own resonance point according to its channel phase
- No boundary pile-up (except R) $\Rightarrow$ the chosen bounds **do not strangle the solution**; the varactor range is used almost entirely – validating the datasheet-based choice



- 1 Context and final problem statement
- 2 Physical constraints and the origin of every bound
- 3 Circuit-model verification
- 4 Algorithm design: from plain PSO to memetic PSO
- 5 Simulation results
- 6 Complexity and conclusion**

- One whole-swarm fitness evaluation: $O(n_p N(M + 4))$ operations (decoding, Z_n , v_n , rate)

Total cost of one memetic optimization run

$$C_{\text{memetic}} = \underbrace{O(T n_p N(M + 4))}_{\text{PSO}} + \underbrace{O(T_{\text{LB}} D N(M + 4))}_{\text{L-BFGS-B (finite-difference gradients)}}, \quad T_{\text{LB}} \leq 160$$

- For $N = 40$: $T = 640$, $n_p = 96$ – the PSO stage dominates but is fully NumPy-vectorized
- The full final experiment suite ($\approx 13,500$ optimization runs): **≈ 54 minutes** on a 12-core machine, versus 6–7 hours sequentially

- IRS optimization moved from the *abstract phase* level to the **physical component** level $\{L_{1,n}, L_{2,n}, C_n, R_n\}$; circuit chain verified independently ($\beta_{\min}, k, \phi$ reproduced, RMSE 0.0029)
- Every bound traced to the reference paper and 2.4-GHz technology limits
- Five-version journey: adaptive budget $\rightarrow$ four variables $\rightarrow$ warm start $\rightarrow$ memetic L-BFGS-B

Final result (500 realizations, $d = 498$ m, $N = 40$)

PSO over all four components reaches **3.211 bps/Hz = 97.3%** of the ideal upper bound (3.301),
outperforming the paper's AO baseline (2.320) by **38.4%**.

Overarching lesson: PSO delivers its full power only when the *search budget, initialization quality, and local refinement* match the problem's dimensionality; the residual gap to the ideal bound is the unavoidable loss of any real passive circuit.

Thank you!

We are happy to take questions

Electronics for IT | MiniProject20252 | 2026